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Semiconductor device

Abstract

A semiconductor device, including, a drift region of a first conductivity type provided on a semiconductor substrate; a field stop region of a first conductivity type provided below the drift region and having one or more peaks; and a collector region of a second conductivity type provided below the field stop region, wherein when an integral concentration of the collector region is set to be x [cm.sup.−2], a depth of a first peak that is a shallowest from the back surface of the semiconductor substrate out of the one or more peaks is set to be y**1** [μm], line A**1**: y**1**=(−7.4699E−01)ln(x)+(2.7810E+01), and line B**1**: y**1**=(−4.7772E−01)ln(x)+(1.7960E+01), a depth of the first peak and the integral concentration are within a range between a line A**1** and a line B**1**, is provided.

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Background/Summary

(1) The contents of the following Japanese patent application(s) are incorporated herein by reference: NO. 2020-121285 filed in JP on Jul. 15, 2020, and NO. PCT/JP2021/026375 filed in WO on Jul. 13, 2021.
BACKGROUND
1. Technical Field
(2) The present invention relates to a semiconductor device.
2. Related Art
(3) Conventionally, a semiconductor device including a field stop region is known (for example, see Patent Document 1).
(4) Patent Document 1: Japanese Patent Application Publication No. 2015-135954

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is an example of a top view of a semiconductor device **100**.
(2) FIG. **1**B is an example of a top view of the semiconductor device **100** which corresponds to the region A of FIG. **1**A.
(3) FIG. **1**C is a diagram showing an example in the cross section b-b′ of FIG. **1**B.
(4) FIG. **1**D is a diagram showing an example of a doping concentration distribution in the depth direction in the position of the line c-c′ in FIG. **1**C.
(5) FIG. **1**E is an enlarged view of a field stop region **20** and a collector region **22** in FIG. **1**D.
(6) FIG. **1**F is a diagram showing an example of distributions of a net doping concentration in the depth direction and a hydrogen chemical concentration in a first concentration peak and the collector region **22**.
(7) FIG. **1**G is an example of a top view of the semiconductor device **100** which corresponds to the region B of FIG. **1**A.
(8) FIG. **1**H is a diagram showing an example in the cross section a-a′ of FIG. **1**A.
(9) FIG. **2**A is an example of a circuit upon a clamping overload tolerance test of the semiconductor device **100**.
(10) FIG. **2**B is a diagram for describing clamping energy of the semiconductor device **100**.

(11) FIG. **3**A shows a relation between the doping concentration of the collector region **22** and clamping energy.
(12) FIG. **3**B shows a relation between the doping concentration of the collector region **22** and the breakdown voltage of an element.
(13) FIG. **4**A shows an example of the current/voltage characteristic of the semiconductor device **100** according to the embodiment example.
(14) FIG. **4**B shows an example of the current/voltage characteristic of the semiconductor device **500** according to the comparative example.
(15) FIG. **5**A shows a relation between the integral concentration of the collector region **22** and the depth of a first peak P**1**.
(16) FIG. **5**B shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**.
(17) FIG. **5**C shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**.
(18) FIG. **6**A shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**.
(19) FIG. **6**B shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**.
(20) FIG. **6**C shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**.
(21) FIG. **6**D shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**.
(22) FIG. **6**E shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**.
(23) FIG. **6**F shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**.
(24) FIG. **7**A shows another example of a relation between the integral concentration of the collector region **22** and the depth of a second peak P**2**.
(25) FIG. **7**B shows another example of a relation between the integral concentration of the collector region **22** and the depth of the second peak P**2**.
(26) FIG. **7**C shows another example of a relation between the integral concentration of the collector region **22** and the depth of the second peak P**2**.
(27) FIG. **8**A is a diagram for describing a relation between float inductance Ls and collector current decrease rate dIce/dt.
(28) FIG. **8**B is a diagram for describing a relation between comparative stray inductance Ls.Math.A and current density dJce/dt.
(29) FIG. **9** is a diagram for describing a relation between comparative stray inductance Ls.Math.A and comparative gate resistance Rg.Math.A.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(30) Hereinafter, the present invention will be described through embodiments of the invention, but the following embodiments do not limit the claimed invention. In addition, not all combinations of features described in the embodiments are essential to the solution of the invention.
(31) In the present specification, one side in a direction parallel to the depth direction of a semiconductor substrate is referred to as an “upper” side, and the other side is referred to as a “lower” side. One surface of two principal surfaces of a substrate, a layer or other member is referred to as an upper surface, and the other surface is referred to as a lower surface. The “upper”, “lower”, “front”, and “back” directions are not limited to the gravitational direction or the direction of attachment to a substrate or the like at the time of mounting of a semiconductor device.
(32) In the present specification, technical matters may be described using orthogonal coordinate axes of an X axis, a Y axis, and a Z axis. The orthogonal coordinate axes merely specify relative

positions of components, and do not limit a specific direction. For example, the Z axis is not limited to indicate the height direction with respect to the ground. Note that a +Z axis direction and a −Z axis direction are directions opposite to each other. When the Z axis direction is described without describing the signs, it means that the direction is parallel to the +Z axis and the −Z axis.
(33) In the present specification, a surface parallel to the upper surface of the semiconductor substrate is referred to as the XY surface, and an orthogonal axis parallel to the upper surface and the lower surface of the semiconductor substrate is referred to as the X axis and the Y axis. Further, an axis perpendicular to the upper surface and the lower surface of the semiconductor substrate is referred to as the Z axis. The depth direction of a semiconductor substrate may be referred to as the Z axis. Additionally, as used in the present specification, the view of the semiconductor substrate in the Z axis direction is referred to as a planar view. Further, in the present specification, a direction parallel to the upper surface and the lower surface of the semiconductor substrate may be referred to as a horizontal direction, including an X axis direction and a Y axis direction.
(34) In each embodiment example, an example in which a first conductivity type is N type and a second conductivity type is P type is described. However, the first conductivity type may be P type and the second conductivity type may be N type. In this case, conductivity types of substrates, layers, regions, or the like in each embodiment example are of respective opposite polarity.
(35) In the present specification, a case where a term such as “identical” or “equal” is mentioned may include a case where an error due to a variation in manufacturing or the like is included. The error is, for example, within 10%.
(36) In the present specification, a conductivity type of doping region where doping has been carried out with an impurity is described as a P type or an N type. In the present specification, the impurity may particularly mean either a donor of the N type or an acceptor of the P type, and may be described as a dopant. In the present specification, doping means introducing the donor or the acceptor into the semiconductor substrate and turning it into a semiconductor presenting a conductivity type of the N type, or a semiconductor presenting conductivity type of the P type.
(37) In the present specification, a doping concentration means a concentration of the donor or a concentration of the acceptor in a thermal equilibrium state. In the present specification, a net doping concentration means a net concentration obtained by adding the donor concentration set to be a positive ion concentration to the acceptor concentration set to be a negative ion concentration, taking into account of polarities of charges. As an example, when the donor concentration is N.sub.D and the acceptor concentration is N.sub.A, the net net doping concentration at any position is given as N.sub.D−N.sub.A.
(38) The donor has a function of supplying electrons to a semiconductor. The acceptor has a function of receiving electrons from the semiconductor. The donor and the acceptor are not limited to the impurities themselves. For example, a VOH defect which is a combination of a vacancy (V), oxygen (O), and hydrogen (H) existing in the semiconductor functions as the donor that supplies electrons. A VOH defect may be referred to simply as a hydrogen donor.
(39) In the present specification, a description of a P+ type or an N+ type means a higher doping concentration than that of the P type or the N type, and a description of a P− type or an N− type means a lower doping concentration than that of the P type or the N type. Further, in the present specification, a description of a P++ type or an N++ type means a higher doping concentration than that of the P+ type or the N+ type.
(40) A chemical concentration in the present specification indicates an atomic density of an impurity measured regardless of an electrical activation state. The chemical concentration can be measured by, for example, secondary ion mass spectrometry (SIMS). The net doping concentration described above can be measured by voltage-capacitance profiling (CV profiling). Further, a carrier concentration measured by spreading resistance profiling (srp method) may be set to be the net doping concentration. The carrier concentration measured by the CV profiling or the srp method may be a value in a thermal equilibrium state. Further, in a region of the N type, the donor

concentration is sufficiently greater than the acceptor concentration, and thus the carrier concentration of the region may be set to be the donor concentration. Similarly, in a region of the P type, the carrier concentration of the region may be set to be the acceptor concentration. Each concentration in the present invention may be a value at room temperature. For the value at room temperature, a value at 300 K (Kelvin) (about 26.9° C.) may be used as an example.

(41) Further, when a concentration distribution of the donor, acceptor, or net doping has a peak in a region, the peak value may be set to be the concentration of the donor, acceptor, or net doping in the region. In a case where the concentration of the donor, acceptor or net doping is substantially uniform in a region, or the like, an average value of the concentration of the donor, acceptor or net doping in the region may be set to be the concentration of the donor, acceptor or net doping.

(42) The carrier concentration measured by the srp method may be lower than the concentration of the donor or the acceptor. In a range where a current flows when a spreading resistance is measured, carrier mobility of the semiconductor substrate may be lower than a value in a crystalline state. The reduction in carrier mobility occurs when carriers are scattered due to disorder (disorder) of a crystal structure due to a lattice defect or the like.

(43) The concentration of the donor or the acceptor calculated from the carrier concentration measured by the CV profiling or the srp method may be lower than a chemical concentration of an element indicating the donor or the acceptor. As an example, in a silicon semiconductor, a donor concentration of phosphorous or arsenic serving as a donor, or an acceptor concentration of boron serving as an acceptor is approximately 99% of chemical concentrations of these. On the other hand, in a silicon semiconductor, a donor concentration of hydrogen serving as a donor is, for example, approximately from 0.1% to 10% of chemical concentration of hydrogen.

(44) Unless otherwise stated, SI unit system is used as unit system herein. Although a unit of a length may be indicated in cm or the like, various calculations may be performed after converting the unit into meters (m).

(45) As used herein, a doping concentration refers to a concentration of a donor or an acceptorized dopant. Therefore, the unit is /cm.sup.3. As used herein, a difference of concentrations of a donor and an acceptor (i.e., a net doping concentration) may be set to be the doping concentration. In this case, the doping concentration may be measured by an srp method. Moreover, a chemical concentration of the donor and the acceptor may also be set to be a doping concentration. In this case, the doping concentration can be measured by an SIMS method. Unless otherwise limited, any one of the above may be used as a doping concentration. Unless otherwise limited, a peak value of a doping concentration distribution in a doping region may be set to be a doping concentration in the doping region.

(46) In addition, as used herein, a dosage refers to the number of ions implanted in a wafer per unit area when ions are implanted. Therefore, the unit is /cm.sup.2. Note that a dosage of a semiconductor region can be taken as an integral concentration which is obtained by integrating doping concentrations across the semiconductor region in the depth direction. The unit of the integral concentration is /cm.sup.2. Therefore, the dosage may be treated as the same as the integral concentration. The integral concentration may also be set to be an integral value within a half-value width. In the case of being overlapped by spectrum of another semiconductor region, the integral concentration may be derived without the influence of another semiconductor region.

(47) Therefore, as used in the present specification, the level of the doping concentration can be read as the level of the dosage. That is, if the doping concentration of one region is higher than the doping concentration of another region, it can be understood that the dosage of the one region is higher than the dosage of the another region.

(48) FIG. **1**A is an example of a top view of a semiconductor device **100**. The semiconductor device **100** includes an active region **110**, an outer peripheral region **120**, and a pad region **130**. The semiconductor device **100** is a semiconductor chip including a transistor portion **70** and a diode portion **80**. The semiconductor device **100** includes a temperature sensing portion **76**, and may be

mounted on a module such as IPM (intelligent power module).
(49) The transistor portion **70** includes a transistor such as an IGBT (Insulated Gate Bipolar Transistor). The diode portion **80** includes a diode such as a free wheel diode (FWD). The semiconductor device **100** of the present example is a reverse conducting IGBT (RC-IGBT) having a transistor portion **70** and a diode portion **80** on the same chip. In the present example, the transistor of the transistor portion **70** is an IGBT.
(50) The semiconductor substrate **10** may be a silicon substrate, a silicon carbide substrate, or a nitride semiconductor substrate or the like such as gallium nitride. The semiconductor substrate **10** in the present example is a silicon substrate. The semiconductor substrate **10** has an active region **110** and an outer peripheral region **120**.
(51) The transistor portion **70** and the diode portion **80** may be arranged alternately in a cyclical manner on the XY plane. The semiconductor device **100** of the present example includes a plurality of transistor portions **70** and a plurality of diode portions **80** respectively. The transistor portion **70** and the diode portion **80** of the present example have trench portions that extend in the Y axis direction. However, the transistor portion **70** and the diode portion **80** may have trench portions that extend in the X axis direction.
(52) A temperature sensing portion **76** is provided above an active region **110**. The temperature sensing portion **76** may be formed of monocrystalline or polycrystalline silicon. When the temperature of the semiconductor device **100** changes, the forward voltage of a current flowing in the temperature sensing portion **76** changes. Based on the change of the forward voltage, the temperature of the semiconductor device **100** can be detected.
(53) The active region **110** has the transistor portion **70** and the diode portion **80**. The active region **110** is a region where the main current flows between the front surface and the back surface of the semiconductor substrate **10** when the semiconductor device **100** is controlled to be an ON-state. That is, it is a region in which the current flows through the interior of the semiconductor substrate **10** from the front surface to the back surface or from the back surface to the front surface of the semiconductor substrate **10** in the depth direction. In the present specification, the transistor portions **70** and the diode portions **80** may be referred to as an element portion or an element region respectively.
(54) Moreover, in a top view, a region sandwiched by two element portions is referred to as an active region **110**. In the present example, a region where a gate runner **51** is provided being sandwiched by element portions is included in the active region **110**.
(55) The gate runner **51** may supply gate potential supplied from a gate pad **131** of a pad region **130**, to a gate conductive portion of the transistor portion **70**. The gate runner **51** is, in a top view, provided along the outer periphery of the active region **110**. The gate runner **51** may be, in a top view, provided in the region between the transistor portion **70** and the diode portion **80**.
(56) The outer peripheral region **120** is, in a top view, provided around an end portion of the semiconductor substrate **10** so as to enclose the active region **110** and the pad region **130**. It is a region between the active region **110** and the pad region **130** and the outer peripheral edge of the semiconductor substrate **10**. The outer peripheral region **120** is, in a top view, provided around the active region **110**. The outer peripheral region **120** may have an edge termination structure portion. The edge termination structure portion reduces electric field strength at the side of the front surface of the semiconductor substrate **10**. For example, the edge termination structure portion has a structure of a guard ring, a field plate, an RESURF, and a combination thereof.
(57) The pad region **130** has a gate pad **131**, a sense-IGBT **132**, a sense-emitter pad **133**, an anode pad **134**, and a cathode pad **135**. In the present example, a cathode pad **135**, an anode pad **134**, a gate pad **131**, a sense-IGBT **132**, and a sense-emitter pad **133** are provided in a row in the X axis direction in this order. Each pad may be an electrode pad including gold (Au), silver (Ag), copper (Cu), aluminum (Al), or the like.
(58) The gate pad **131** is electrically connected to a gate conductive portion of the transistor portion

70 via a gate runner **51**. The gate pad **131** is set to a gate potential.
(59) The sense-IGBT **132** is an IGBT for detecting a main current that flows in the transistor portion **70**. A current proportional to a main current that flows in the transistor portion **70** flows in the sense-IGBT **132**. By taking the sense-current flowing in the sense-IGBT **132** into a control circuit provided in the exterior to the semiconductor device **100** allows the main current of the transistor portion **70** to be detected. The sense-emitter pad **133** of the present example has the same potential as the emitter of the sense-IGBT **132**. The sense-current may be taken into the above-described control circuit from the sense-emitter pad **133** via the sense-IGBT **132**. The control circuit may sense the main current based on the sense-current so as to shut down a current flowing in the transistor portion **70** when a overcurrent is flowing in the transistor portion **70**.
(60) An anode pad **134** is electrically connected to the temperature sensing portion **76**, and set to the anode potential of the temperature sensing portion **76**. Similarly, a cathode pad **135** is electrically connected to the temperature sensing portion **76**, and set to the cathode potential of the temperature sensing portion **76**. The use of the anode pad **134** and the cathode pad **135** allows potential differences between the anode and the cathode of the temperature sensing portion **76** to be sensed.
(61) FIG. **1**B is an example of a top view of the semiconductor device **100** which corresponds to the region A of FIG. **1**A. That is, FIG. **1**B shows an enlarged view of the end portion of the active region **110**.
(62) The transistor portion **70** is a region where the collector region **22** provided on the back surface side of the semiconductor substrate **10** is projected onto a front surface of the semiconductor substrate **10**. The collector region **22** has the second conductivity type. The collector region **22** of the present example is of P+ type, by way of example. The transistor portion **70** includes a boundary portion **90** that is positioned in a boundary between the transistor portion **70** and the diode portion **80**.
(63) The diode portion **80** is a region where the cathode region **82** provided on the back surface side of the semiconductor substrate **10** is projected onto the front surface of the semiconductor substrate **10**. The cathode region **82** has the first conductivity type. The cathode region **82** of the present example is of N+ type, by way of example.
(64) The semiconductor device **100** in the present example includes a gate trench portion **40**, a dummy trench portion **30**, an emitter region **12**, a base region **14**, a contact region **15**, and a well region **17**, in the front surface of the semiconductor substrate **10**. In addition, the semiconductor device **100** of the present example includes an emitter electrode **52** and a gate metal layer **50** provided above the front surface of the semiconductor substrate **10**.
(65) The emitter electrode **52** is provided above the gate trench portion **40**, the dummy trench portion **30**, the emitter region **12**, the base region **14**, the contact region **15** and the well region **17**. Also, the gate metal layer **50** is provided above the well region **17** and the gate runner **51**. The emitter electrode **52** of the present example is set to an emitter potential of the transistor portion **70**.
(66) The gate metal layer **50** is electrically connected to the gate conductive portion of the transistor portion **70** and supplies a gate voltage to the transistor portion **70**. The gate metal layer **50** is electrically connected with the gate pad **131**. The gate metal layer **50** is, in a top view, provided along the outer periphery of the active region **110**. The gate metal layer **50** may be, in a top view, provided in the region between the transistor portion **70** and the diode portion **80**.
(67) The gate runner **51** connects the gate metal layer **50** with a gate conductive portion within the gate trench portion **40**. The gate runner **51** is not connected to a dummy conductive portion in the dummy trench portion **30**. For example, the gate runner **51** is formed of polysilicon doped with impurities, or the like.
(68) The emitter electrode **52** and the gate metal layer **50** are formed of a material including metal. For example, at least partial region of the emitter electrode **52** may be formed of aluminum, aluminum-silicon alloy, or aluminum-silicon-copper alloy. The emitter electrode **52** may include

barrier metal formed of titanium, titanium compound, and the like under the region formed of aluminum and the like. The emitter electrode **52** and the gate metal layer **50** are provided separately from each other.
(69) The emitter electrode **52** and the gate metal layer **50** are provided above the semiconductor substrate **10** with an interlayer dielectric film **38** interposed therebetween. The interlayer dielectric film **38** is omitted in FIG. **1**B. A contact hole **54**, a contact hole **55**, and a contact hole **56** are provided through the interlayer dielectric film **38**.
(70) The contact hole **55** connects the gate metal layer **50** with the gate runner **51**. Inside the contact hole **55**, a plug formed of tungsten or the like may be formed. The contact hole **55** may be provided along the gate runner **51**.
(71) The contact hole **56** connects the emitter electrode **52** with a dummy conductive portion in a dummy trench portion **30**. Inside the contact hole **56**, a plug formed of tungsten or the like may be formed.
(72) A connection portion **25** electrically connects the emitter electrode **52** with a plug in the interior of the contact hole **56**. The connection portion **25** has a conductive material such as polysilicon doped with impurities. The connection portion **25** of the present example is a polysilicon layer doped with impurities of the N type. The connection portion **25**, in a top view, covers a range larger than the contact hole **56**. The connection portion **25** is provided above the front surface of the semiconductor substrate **10** via a dielectric film such as an oxide film, or the like.
(73) The gate trench portion **40** is arranged at a predefined interval along a predefined arrangement direction (the X axis direction in the present example). The gate trench portion **40** in the present example may have two extending portions **41** extending along the extending direction (in the present example, the Y axis direction) that is parallel to the front surface of the semiconductor substrate **10** and perpendicular to the arrangement direction and a connection section **43** connecting the two extending portions **41**.
(74) At least a part of the connection section **43** is preferably formed in a curved shape. By connecting end portions of the two extending portions **41** of the gate trench portions **40**, an electric field strength at the end portions of the extending portions **41** can be reduced. In the connection section **43** of the gate trench portion **40**, the gate runner **51** may be connected with the gate conductive portion.
(75) The dummy trench portion **30** is a trench portion electrically connected with the emitter electrode **52**. The dummy trench portion **30** is arranged, similarly to the gate trench portion **40**, at a predefined interval along a predefined arrangement direction (the X axis direction in the present example). The dummy trench portion **30** of the present example may have, similarly to the gate trench portion **40**, a U shape on the front surface of the semiconductor substrate **10**. That is, the dummy trench portion **30** may have two extending portions **31** extending along the extending direction and a connection section **33** connecting the two extending portions **31**.
(76) The well region **17** is a region of a second conductivity type provided closer to the front surface of the semiconductor substrate **10** than the drift region **18**, which will be described below. The well region **17** is an example of the well region provided on the edge side of the semiconductor device **100**. The well region **17** is of P+ type, by way of example. The well region **17** is formed within a predefined range from the end portion of the active region **110** of the side provided with the gate metal layer **50**. A diffusion depth of the well region **17** may be deeper than depths of the gate trench portion **40** and the dummy trench portion **30**. Partial regions of the gate trench portion **40** and the dummy trench portion **30** on the gate metal layer **50** side are formed in the well region **17**. The bottoms at the end of the extending direction of the gate trench portion **40** and the dummy trench portion **30** may be covered by the well region **17**.
(77) The contact hole **54** is formed above each region of the emitter region **12** and the contact region **15** in the transistor portion **70**. Also, the contact hole **54** is provided above the base region

14 in the diode portion **80**. The contact hole **54** is provided above the contact region **15** in the boundary portion **90**. No contact hole **54** is provided above the well region **17** provided on the both ends in the extending direction. In this manner, one or more contact holes **54** are formed in the interlayer dielectric film **38**. One or more contact holes **54** may be provided to extend in the extending direction.
(78) The boundary portion **90** is a region provided in the transistor portion **70** and adjacent to the diode portion **80**. The boundary portion **90** includes the contact region **15**. The boundary portion **90** of the present example does not include the emitter region **12**. The boundary portion **90** of the present example is arranged so that both ends in the arrangement direction become dummy trench portions **30**.
(79) The mesa portion **71**, the mesa portion **91** and the mesa portion **81** are mesa portions provided adjacent to the trench portions, in a plane parallel to the front surface of the semiconductor substrate **10**. The mesa portion may be a portion of the semiconductor substrate **10** sandwiched between two adjacent trench portions, which is located from the front surface of the semiconductor substrate **10** to the depth of the deepest bottom portion of each trench portion. The extending portions of each trench portion may be set to be one trench portion. That is, the region sandwiched between two extending portions may be set to be a mesa portion.
(80) The mesa portion **71** is provided adjacent to at least one of the dummy trench portion **30** or the gate trench portion **40** in the transistor portion **70**. The mesa portion **71** has the well region **17**, the emitter region **12**, the base region **14**, and the contact region **15**, in the front surface of the semiconductor substrate **10**. In the mesa portion **71**, the emitter region **12** and the contact region **15** are provided alternately in the extending direction.
(81) The mesa portion **91** is provided in the boundary portion **90**. The mesa portion **91** has, on the front surface of the semiconductor substrate **10**, a base region **14**, a contact region **15**, and a well region **17**. Although the mesa portion **91** of the present example has both ends in the arrangement direction in contact with the dummy trench portion **30**, at least one may be in contact with the gate trench portion **40**. In the present example, although one mesa portion **91** is provided, a plurality of mesa portions **91** may be provided.
(82) The mesa portion **81** is provided in a region interposed between the dummy trench portions **30** adjacent to each other in the diode portion **80**. The mesa portion **81** has, on the front surface of the semiconductor substrate **10**, a base region **14**, and a well region **17**.
(83) The base region **14** is a region of second conductivity type provided in the transistor portion **70** and the diode portion **80** on the side of the front surface of the semiconductor substrate **10**. The base region **14** is of the P− type, by way of example. The base region **14** may be, in the front surface of the semiconductor substrate **10**, provided in both end portions in the extending direction of the mesa portion **71** and the mesa portion **91**. Moreover, FIG. **1**B shows only one end portion in the extending direction of the base region **14**.
(84) The emitter region **12** is a region of the first conductivity type having a higher doping concentration than that of the drift region **18**. The emitter region **12** of the present example is of N+ type, by way of example. An example of a dopant of the emitter region **12** is arsenic (As). The emitter region **12** is provided in contact with the gate trench portion **40** in the front surface of the mesa portion **71**. The emitter region **12** may be provided being extended in the arrangement direction from one trench portion to the other trench portion of two trench portions that sandwich the mesa portion **71**. The emitter region **12** is also provided below the contact hole **54**.
(85) In addition, the emitter region **12** may or may not be in contact with the dummy trench portion **30**. The emitter region **12** in the present example is in contact with the dummy trench portion **30**. The emitter region **12** may not be provided in the mesa portion **91** of the boundary portion **90**.
(86) The contact region **15** is a region of the second conductivity type having a higher doping concentration than that of the base region **14**. The contact region **15** of the present example is of P+ type, by way of example. The contact region **15** of the present example is provided on the front

surfaces of the mesa portion **71** and the mesa portion **91**. The contact region **15** may be provided in the arrangement direction from one trench portion to the other trench portion of two trench portions that sandwich the mesa portion **71** or the mesa portion **91**. The contact region **15** may be or may not be in contact with the gate trench portion **40**. Moreover, the contact region **15** may be or may not be in contact with the dummy trench portion **30**. In the present example, the contact region **15** is in contact with the dummy trench portion **30** and the gate trench portion **40**. The contact region **15** is also provided below the contact hole **54**. Moreover, the contact region **15** may also be provided in the mesa portion **81**.

(87) FIG. **1**C is a diagram showing an example in the cross section b-b′ of FIG. **1**B. The cross section b-b′ is, in the transistor portion **70**, an XZ surface that passes through the emitter region **12**. The semiconductor device **100** of the present example includes, in the cross section b-b′, the semiconductor substrate **10**, the interlayer dielectric film **38**, the emitter electrode **52**, and the collector electrode **24**. The emitter electrode **52** is formed above the semiconductor substrate **10** and the interlayer dielectric film **38**.

(88) The drift region **18** is a region of the first conductivity type provided in the semiconductor substrate **10**. The drift region **18** of the present example is of N− type, by way of example. The drift region **18** may be a remaining region where another doping region is not formed in the semiconductor substrate **10**. That is, the doping concentration of the drift region **18** may be a doping concentration of the semiconductor substrate **10**.

(89) The field stop region **20** is a region of the first conductivity type provided below the drift region **18**. The field stop region **20** of the present example is of N type, by way of example. The doping concentration of the field stop region **20** is higher than a doping concentration of the drift region **18**. The field stop region **20** prevents a depletion layer spreading from the side of the lower surface of the base region **14** from reaching to the collector region **22** of the second conductivity type and the cathode region **82** of the first conductivity type.

(90) The field stop region **20** may have one or more peaks. The field stop region **20** of the present example has four peaks of the first peak P**1** to the fourth peak P**4**. The dopant of one or more peaks may be hydrogen.

(91) The first peak P**1** to the fourth peak P**4** are provided in the order from the back surface **23** in this order. That is, the first peak P**1** is a peak that is the closest to the back surface **23**. The doping concentration of the first peak P**1** may be higher than the doping concentration of the other peaks. Thereby, a depletion layer when a voltage is applied can be slowly and surely stopped.

(92) The collector region **22** is, in the transistor portion **70**, provided below the field stop region **20**. The cathode region **82** is, in the diode portion **80**, provided below the field stop region **20**. The boundary between the collector region **22** and the cathode region **82** is a boundary between the transistor portion **70** and the diode portion **80**.

(93) The collector electrode **24** is formed on the back surface **23** of the semiconductor substrate **10**. The collector electrode **24** is formed of a conductive material such as metal.

(94) The base region **14** is a region of the second conductivity type provided above the drift region **18**, in the mesa portion **71**, the mesa portion **91**, and the mesa portion **81**. The base region **14** is provided in contact with gate trench portion **40**. The base region **14** may be provided in contact with the dummy trench portion **30**.

(95) The emitter region **12** is, in the mesa portion **71**, provided between the base region **14** and the front surface **21**. The emitter region **12** is provided in contact with the gate trench portion **40**. The emitter region **12** may be or may not be in contact with the dummy trench portion **30**. Moreover, the emitter region **12** may not be provided in the mesa portion **91**.

(96) The contact region **15** is provided above the base region **14** in the mesa portion **91**. The contact region **15** is provided in contact with the dummy trench portion **30** in the mesa portion **91**. In another cross section, the contact region **15** may be provided on the front surface **21** in the mesa portion **71**.

(97) An accumulation region **16** is a region of the first conductivity type provided closer to the front surface **21** of the semiconductor substrate **10** than the drift region **18**. The accumulation region **16** of the present example is of N+ type, by way of example. The accumulation region **16** is provided in the transistor portion **70** and the diode portion **80**. The accumulation region **16** of the present example is also provided in the boundary portion **90**.
(98) In addition, the accumulation region **16** is provided in contact with the gate trench portion **40**. The accumulation region **16** may be or may not be in contact with the dummy trench portion **30**. The doping concentration of the accumulation region **16** is higher than the doping concentration of the drift region **18**. Providing the accumulation region **16** can enhance the carrier injection enhancement effect (IE effect) to reduce the ON voltage of the transistor portion **70**.
(99) One or more gate trench portions **40** and one or more dummy trench portions **30** are provided on the front surface **21**. Each trench portion is provided from the front surface **21** to the drift region **18**. In the region where at least any one of the emitter region **12**, the base region **14**, the contact region **15**, and the accumulation region **16** is provided, each trench portion also penetrates these regions to reach the drift region **18**. The configuration of the trench portion penetrating the doping region is not limited to the one manufactured in the order of forming the doping region and then forming the trench portion. The configuration of the trench portion penetrating the doping region also includes a configuration of the doping region being formed between the trench portions after forming the trench portion.
(100) The gate trench portion **40** has a gate trench, a gate dielectric film **42**, and a gate conductive portion **44** that are formed in the front surface **21**. The gate dielectric film **42** is formed to cover an inner wall of the gate trench. The gate dielectric film **42** may be formed by oxidizing or nitriding a semiconductor on the inner wall of the gate trench. The gate dielectric film **42** is formed in the interior of the gate trench, and the gate conductive portion **44** is formed inside the gate dielectric film **42**. The gate dielectric film **42** insulates the gate conductive portion **44** from the semiconductor substrate **10**. The gate conductive portion **44** is formed of a conductive material such as polysilicon. The gate trench portion **40** is covered by the interlayer dielectric film **38** in the front surface **21**.
(101) The gate conductive portion **44** includes a region opposing the adjacent base region **14** in the mesa portion **71** side by sandwiching the gate dielectric film **42**, in the depth direction of the semiconductor substrate **10**. When a predefined voltage is applied to the gate conductive portion **44**, a channel as an inversion layer of electrons is formed in the interfacial surface layer of the base region **14** that is in contact with the gate trench.
(102) The dummy trench portion **30** may have the identical structure as that of the gate trench portion **40**. The dummy trench portion **30** has a dummy trench, a dummy dielectric film **32** and a dummy conductive portion **34** that are formed on the side of the front surface **21**. The dummy dielectric film **32** is formed covering the inner walls of the dummy trench. The dummy conductive portion **34** is formed in the interior of the dummy trench and formed inside the dummy dielectric film **32**. The dummy dielectric film **32** insulates the dummy conductive portion **34** from the semiconductor substrate **10**. The dummy trench portion **30** is covered by the interlayer dielectric film **38** in the front surface **21**.
(103) The interlayer dielectric film **38** is provided in the front surface **21**. The emitter electrode **52** is provided above the interlayer dielectric film **38**. In the interlayer dielectric film **38**, one or more contact holes **54** are provided for electrically connecting the emitter electrode **52** with the semiconductor substrate **10**. The contact hole **55** and the contact hole **56** may also be provided extending through the interlayer dielectric film **38**.
(104) FIG. **1**D is a diagram showing an example of a doping concentration distribution in the depth direction in the position of the line c-c′ in FIG. **1**C. The line c-c′, in the transistor portion **70**, passes through from the emitter region **12** to the collector region **22**. The vertical axis in FIG. **1**D is the logarithmic axis. Moreover, although, in the present example, a doping concentration distribution

of the field stop region **20** in the transistor portion **70** is described, the field stop region **20** in the diode portion **80** may also have similar doping concentration distribution.

(105) A doping concentration of the drift region **18** of the present example is a bulk donor concentration Db. A bulk donor of the first conductivity type (N type) is distributed throughout the semiconductor substrate **10** of the present example. The bulk donor is a dopant donor substantially uniformly contained in an ingot during the manufacture of the ingot from which the semiconductor substrate **10** is made. The bulk donor of the present example is an element other than hydrogen. Although the dopant of the bulk donor is, for example, phosphorous, antimony, arsenic, selenium, and sulfur, it is not limited to these. The bulk donor in the present example is phosphorous. The bulk donor is also contained in the P type region. The semiconductor substrate **10** may be a wafer cut out from an ingot of a semiconductor, or may be a chip made from making a wafer into an individual piece. The semiconductor ingot may be manufactured by either a Chokralsky method (CZ method), a magnetic field applied Chokralsky method (MCZ method), or a float zone method (FZ method). The ingot in the present example is manufactured by the MCZ method. For a bulk donor concentration Db, a chemical concentration of a donor that is distributed throughout the semiconductor substrate **10** may be used, and it may be a value between from 90% to 100% of the chemical concentration.

(106) In the present example, a case where a doping concentration distribution in the field stop region **20** has four concentration peaks P**1**, P**2**, P**3**, P**4** in different positions in the depth direction is described. However, the number of the concentration peaks is not limited to this. A concentration peak in the present example is a peak of donor concentration. A plurality of concentration peaks can be formed by implanting impurities such as hydrogen or phosphorous into a plurality of depth positions of the field stop region **20**. As an example, the position of P**1** is set to be phosphorous, and for the P**2** to P**4**, hydrogen as donors may be used. In the present example, for all of P**1** to P**4** hydrogen as donors are used. The field stop region **20** may have concentration peaks of impurities such as hydrogen or phosphorous at the position corresponding to a concentration peak. The concentration peak of impurities is a peak in the chemical concentration distribution of impurities. Providing the plurality of concentration peaks can suppress the depletion layer from reaching the collector region **22** better.

(107) FIG. **1**E is an enlarged view of a doping concentration distribution in the field stop region **20** and a collector region **22** of FIG. **1**C. The vertical axis in FIG. **1**E is the logarithmic axis. In FIG. **1**E, the peak values of doping concentrations of a plurality of concentration peaks P**1**, P**2**, P**3**, P**4** are d**1**, d**2**, d**3**, d**4** respectively. Also, the depth of pn junction between the collector region **22** and the field stop region **20** is set to be J**1**.

(108) A plurality of concentration peaks includes shallowest peak closest to the back surface **23** of the semiconductor substrate **10**. In the present example, the concentration peak P**1** corresponds to shallowest peak. The concentration peak P**1** of the present example is a concentration peak closest to the collector region **22**. In the field stop region **20** of the diode portion **80**, the concentration peak P**1** is a concentration peak that is the closest to the cathode region **82**. The cathode region **82** may be formed by implanting impurities different from concentration peaks. For example, the cathode region **82** has an impurity concentration peak such as that of phosphorous, and the field stop region **20** has concentration peaks of impurities such as that of hydrogen.

(109) A plurality of concentration peaks includes high concentration peaks arranged in positions that are apart from the back surface **23** than the shallowest peak (the concentration peak P**1**). The high concentration peak may be a concentration peak P**2** closest to the shallowest peak, or may be another concentration peak. In the example of FIG. **1**E, the concentration peak P**2** closest to the concentration peak P**1** corresponds to the high concentration peak.

(110) A plurality of concentration peaks includes low concentration peaks arranged in positions that are apart from the back surface **23** than the high concentration peaks, and whose peak values of doping concentrations are 1/5 or less of peak values of the high concentration peaks. The low

concentration peak may be the deepest peak (in the present example, the concentration peak P**4**) arranged the furthest from the back surface **23** among a plurality of concentration peaks. The low concentration peak may be a concentration peak other than the deepest peak. That is, the low concentration peak may be a concentration peak between the high concentration peaks and the deepest peak.

(111) Also, two or more low concentration peaks may be provided. The low concentration peak is preferably arranged next to each other in the depth direction. Among a plurality of concentration peaks, two or more concentration peaks arranged furthest from the back surface **23** may be low concentration peaks. In the example of FIG. **1**E, the concentration peak P**4** that is the deepest peak, the concentration peak P**3** provided in a position that is the next furthest after the deepest peak toward the back surface **23** from the back surface **23**, and the concentration peak P**2** that is the next closest after the P**3** to the back surface **23** are low concentration peaks. A concentration peak other than the concentration peak that is the closest to the back surface **23** may be set to be a low concentration peak. In the example of FIG. **1**E, either the peak value d**3** of a doping concentration of the concentration peak P**3** or the peak value d**4** of a doping concentration of the concentration peak P**4** is 1/5 or less of the peak value d**1** of a doping concentration of the concentration peak P**1**. The concentration peak d**2** to d**4** of the low concentration peaks P**2** to P**4** may be 1/10 or less of the peak value d**1** of the concentration peak P**1**. Moreover, in the present example, the peak value d**3** of P**3** is lower than the peak value d**4** of P**4**. The peak value d**3** of P**3** may be higher than the peak value d**4** of P**4**. That is, peak values from P**2** to P**4** may fall toward the front surface **21**.

(112) In a short circuit state or the like where two semiconductor devices **100** connected in series get switched to an ON-state at the same time, a high-voltage may be applied between an emitter and a collector of the semiconductor devices **100**. In this case, electric fields tend to concentrate in the vicinity of the deepest peak (in the present example, the concentration peak P**4**) in the field stop region **20**. Therefore, if a doping concentration in the vicinity of the deepest peak is raised as in the concentration peak P**3** and the concentration peak P**4**, and the like, it ends up promoting a concentration of electric fields. If electric fields concentrate, the gate voltage tends to vibrate when the semiconductor device **100** turns off or the like.

(113) The concentration peak of the present example provides low concentration peaks whose doping concentration is sufficiently small in a position that is deeper than the high concentration peak (the concentration peak P**1**). Therefore, an electric field strength in deep positions of the field stop region **20** can be reduced. As described above, there may be one or more low concentration peaks, and a plurality of them may be provided. This allows the field stop region **20** with relatively low concentration to be formed elongated on the side of the side of the drift region **18**. In the example of FIG. **1**E, although the field stop region **20** has three low concentration peaks, in another example, the field stop region **20** may have four or more low concentration peaks. Also, the deepest peak may be provided on the side of the front surface **21** of the semiconductor substrate **10**. By forming a low concentration region elongated in the depth direction while making the side of the drift region **18** of the field stop region **20** a low concentration region, it becomes easy to maintain field stop function while reducing electric field strength.

(114) The peak values of doping concentrations of low concentration peaks may be 1/5 or less of peak values of doping concentrations of high concentration peaks, may be 1/10 or less, or may be 1/20 or less. By decreasing doping concentration of low concentration peak, an electric field strength can be reduced more.

(115) Also, the peak values of doping concentrations of low concentration peaks are higher than a bulk donor concentration Db. The peak values of doping concentrations of low concentration peaks may be 50-fold or less of a bulk donor concentration Db of the semiconductor substrate **10**. The doping concentration of the drift region **18** may be used as a bulk donor concentration Db. The peak values of doping concentrations of low concentration peaks may be 20-fold or less, 10-fold or less, 8-fold or less, 5-fold or less, 3-fold or less, or 2-fold or less of a bulk donor concentration Db.

(116) Also, a local minimum value of doping concentration between peaks of low concentration peaks may be higher than a bulk donor concentration Db, or may be 5-fold or less, 3-fold or less, or 2-fold or less of a bulk donor concentration Db. The ratio of local minimum values between peaks for one of the peak concentrations of adjacent low concentration peaks may be 0.8 or less, 0.5 or less, 0.2 or less, 0.1 or more, 0.2 or more, or 0.5 or more.
(117) The positions in the depth direction of the concentration peak P**1**, P**2**, P**3**, P**4** are set to be Z**1**, Z**2**, Z**3**, Z**4** respectively. The distance between the concentration peak P**4** and the concentration peak P**2** in the depth direction is Z**4**-Z**2**. Also, the distance between the concentration peak P**1** and the concentration peak P**2** in the depth direction is Z**2**-Z**1**. The distance Z**4**-Z**2** may be greater than the distance Z**2**-Z**1**. Also, the distance between the concentration peak P**3** and the concentration peak P**4** is Z**4**-Z**3**. The distance between the concentration peak P**3** and the concentration peak P**2** in the depth direction is Z**3**-Z**2**. The distance Z**4**-Z**3** may be greater than the distance Z**3**-Z**2**. Also, the distance Z**4**-Z**3** may be greater than the distance Z**2**-Z**1**.
(118) Also, the peak value of doping concentration of the concentration peak P**4** may be 1/5 or less of peak values of doping concentrations of the high concentration peaks, may be 1/10 or less, or may be 1/20 or less. The peak value of doping concentration of the concentration peak P**4** is higher than a bulk donor concentration Db. The average value of the concentration peak P**4** and the peak value of doping concentration of the concentration peak P**3** may be 50-fold or less, 20-fold or less, 10-fold or less, 8-fold or less, 5-fold or less, 3-fold or less, or 2-fold or less of a bulk donor concentration Db. By making the average value of doping concentration of the two deepest concentration peaks small, the electric field strength in the field stop region **20** of the vicinity of the drift region **18** can be reduced.
(119) In the present example, in the pn junction J**1** between the collector region **22** and the field stop region **20**, a donor concentration or an acceptor concentration may be 1E16/cm.sup.3 or less, 5E15/cm.sup.3 or less, or 2E15/cm.sup.3 or less. Moreover, E means exponentiations of 10, for example, 1E16/cm.sup.3 means 1×10.sup.16/cm.sup.3.
(120) FIG. **1**F is a diagram showing an example of the distributions of a net doping concentration in the depth direction and a hydrogen chemical concentration in the first concentration peak and the collector region **22**. The solid line represents an example of a distribution of a net doping concentration, and the broken line represents an example of a distribution of a hydrogen chemical concentration. The vertical axis in FIG. **1**F is the logarithmic axis. Here, the diagonally shaded regions in FIG. **1**F are an integral concentration of the collector region **22** in the present example. That is, in the present specification, the integral concentration of the collector region **22** means a concentration that results from integrating the net doping concentration of the collector region **22** from the back surface to the pn junction position J**1** of the field stop region **20**, in the depth direction of the semiconductor substrate **10**. Also, the hydrogen chemical concentration Dh in the pn junction J**1** between the collector region **22** and the field stop region **20** may be 1E18/cm.sup.3 or less, 1E17/cm.sup.3 or less, 1E16/cm.sup.3 or more, or 1E15/cm.sup.3 or more.
(121) FIG. **1**G is an example of a top view of the semiconductor device **100** which corresponds to the region B of FIG. **1**A. In the present example, FIG. **1**G shows an enlarged view of the end portion of the active region **110**. The region B is a region that includes the transistor portion **70** and the gate runner **51** of the active region **110**. The semiconductor device **100** of the present example includes the dummy trench region **60** and the well-contact region **65**.
(122) The dummy trench region **60** is a region that has only the dummy trench portion **30** as a trench portion. The dummy trench region **60** is, in the arrangement direction, provided between the gate trench portion **40** that is the closest to the outer peripheral region **120**, and the outer peripheral region **120**. The dummy trench region **60** has a plurality of dummy trench portions **30** spaced by a given interval in the arrangement direction.
(123) The well-contact region **65** is provided in a part of the well region **17**, and the well-contact region **65** extracts a hole injected from the collector region **22** to the emitter electrode **52**. The well-

contact region **65** has the contact regions **15**. The contact hole **54** is provided above the contact region **15** of the well-contact region **65**. The contact region **15** is electrically connected to the emitter electrode **52** via a plurality of contact holes **54**. Moreover, the semiconductor device **100** may not include the well-contact region **65**.

(124) In the mesa portion **61** are provided the base region **14**, the contact region **15**, and the well region **17**. The contact region **15** is, in the mesa portion **61**, provided from one of dummy trench portions **30** adjacent in the arrangement direction to the other dummy trench portions **30**. Because the mesa portion **61** of the present example has a contact region **15**, it becomes easy to extract a hole, when compared to the case without having the contact region **15**. This prevents the destruction of the semiconductor device **100** by the concentration of an avalanche current in the end portion of the well region **17**.

(125) The emitter electrode **52** is provided above the dummy trench region **60** and the well-contact region **65**. The emitter electrode **52** is electrically connected to the front surface **21** of the semiconductor substrate **10** in each of the dummy trench region **60**, the well-contact region **65**, and the transistor portion **70**, via the contact hole **54**.

(126) The well region **17** is, in a top view, provided around the side of an outer periphery of the active region **110**. The inside end portion of the well region **17** is shown by a broken line.

(127) The accumulation region **16** is provided extending from the transistor portion **70** to the dummy trench region **60**. The accumulation region **16** of the present example is provided extending from the transistor portion **70** to the mesa portion **61** at the midway of the dummy trench region **60**. By providing the accumulation region **16** extending from the transistor portion **70** to the dummy trench region **60**, the effect on the accumulation region **16** formed in the transistor portion **70** becomes small even if there happens to be displacements in masks for forming the accumulation region **16**. This can suppress the unevenness of a gate threshold voltage (Vth) and the unevenness of a saturation current. The outer end portion of the accumulation region **16** is shown by a broken line.

(128) FIG. **1**H is a diagram showing an example in the cross section a-a′ of FIG. **1**A. In the present example, a sectional view of a region that overlaps the active region **110** and the outer peripheral region **120** is shown. The outer peripheral region **120** of the present example has a guard ring structure and a channel stopper structure.

(129) A guard ring structure may include a plurality of guard ring portions **92**. The guard ring structure of the present example includes five guard ring portions **92**. Each guard ring portion **92** may be provided so as to surround the active region **110** and the pad region **130** in the front surface **21**.

(130) A guard ring structure may have a function of spreading a depletion layer generated in the active region **110** to outside of the semiconductor substrate **10**. This can prevent an electric field strength inside the semiconductor substrate **10**. Therefore, it is possible to improve the breakdown voltage of the semiconductor device **100**, compared with the case where the guard ring structure is not provided.

(131) The guard ring portion **92** is a semiconductor region of P+ type formed by ion implantation in the vicinity of the front surface **21**. The guard ring portion **92** is electrically connected to the electrode layer **94**. The electrode layer **94** may have the same material as the gate metal layer **50** or the emitter electrode **52**.

(132) A plurality of guard ring portions **92** are electrically insulated with each other by interlayer dielectric films **38**. A depth of the bottom portion of the guard ring portion **92** may be the same depth as the bottom portion of the well region **17**. The depth of the bottom portion of the guard ring portion **92** may be deeper than depths of the gate trench portion **40** and the dummy trench portion **30**.

(133) A channel stopper structure has a channel stopper region **96** and an electrode layer **94**. The channel stopper region **96** is electrically connected to the electrode layer **94** via an opening of the

interlayer dielectric film **38**. The conductivity type of the channel stopper region **96** may be the first conductivity type, or the second conductivity type. The conductivity type of the channel stopper region **96** of the present example is of N+ type. The channel stopper region **96** has a function of terminating a depletion layer generated in the active region **110** at the outer end of the semiconductor substrate **10**.
(134) The well region **17** may go beyond the well-contact region **65** in the arrangement direction so as to further extend to the outside. The well region **17** of the present example may be provided so that the distance between the innermost guard ring portion **92** in the outer peripheral region **120** and the outer end of the well region **17** is close to each other. Moreover, as a modification of the present example, instead of the well region **17**, the base region **14** may be provided by extending it to the vicinity of the innermost guard ring portion **92**. Moreover, above the well region **17**, an oxide film **39** may be provided between the contact region **15** and the gate runner **51**. The oxide film **39** may be formed in the identical process as that of the dummy dielectric film **32** or the gate dielectric film **42**. Alternatively, the oxide film **39** may be formed in a process such as field oxide with thicker film thickness.
(135) FIG. **2**A is an example of a circuit upon a clamping overload tolerance test of the semiconductor device **100**. A gate voltage Vg is imparted on a gate terminal of the semiconductor device **100** via predefined gate resistance Rg. In the clamping overload tolerance test, a rated current Ic is switched by a predefined power source voltage Vcc. The power source voltage Vcc may be approximately 60% of the rated voltage.
(136) The float inductance Ls is a float inductance of a circuit to which the semiconductor device **100** is connected. Because the float inductance Ls is inclined to maintain the current, a voltage overshoot occurs when shutting down the current. By raising the float inductance Ls each switching from a predefined initial value, the semiconductor device **100** is turned off. When the float inductance Ls increases, a snap-off voltage ΔV increases. By increasing the float inductance Ls, the electric field intensity of the interior of the semiconductor device **100** increases so that an element is likely to be disrupted. Here, a result in a test that is one test before the test where the element is disrupted is set to be clamping energy (a breakdown withstand capability).
(137) FIG. **2**B is a diagram for describing clamping energy of the semiconductor device **100**. This figure shows movements of the collector-emitter current Ice at turn-off and the collector-emitter voltage Vce. The graph of the present example shows a waveform immediately prior to the waveform of the float inductance Ls in which an element is disrupted after repeating turn-offs while increasing float inductance Ls in a clamping overload tolerance test. By a turn-off of the semiconductor device **100**, a collector-emitter voltage Vce snaps off and the voltage is kept constant for a certain period of time. The period of time when the voltage is constant is a clamp period.
(138) Clamping energy is an integral value where a current multiplied by a voltage is in the range of 0 or more in the period of time until when the voltage settles from a snap-off of a collector-emitter voltage Vce into a power source voltage Vcc. That is, clamping energy is indicated by an integral value of ∫Ice×Vce dt (i.e., energy value). The greater clamping energy indicates the greater breakdown withstand capability at turn-off.
(139) FIG. **3**A shows a relation between the doping concentration of the collector region **22** and clamping energy. The vertical axis indicates clamping energy [mJ] and the horizontal axis indicates an integral value that integrated the doping concentration of the collector region **22** over the depth from the back surface **23** to the PN junction between the collector region **22** and the field stop region **20** [cm.sup.−2] (hereinafter, referred to as the integral value of the doping concentration of the collector region **22**). A black square indicates the case where the acceleration energy of proton is 400 keV. A white square indicates the case where the acceleration energy of proton is 300 keV. Proton is one kind of hydrogen ions. Hydrogen donor is formed, for example, by the ion implantation of proton into a semiconductor substrate.

(140) The smaller the acceleration energy of proton is, the smaller the total dosage of proton becomes, thereby clamping energy becomes more inclined to rise. Also, when the total dosage of proton is small, concentration of holes injected from the back surface **23** at an avalanche breakdown increases, thereby an avalanche breakdown tends to be generated on the side of active region **110**. In this manner, when the clamping breakdown voltage of active region **110** falls lower than the clamping breakdown voltage of the outer peripheral region **120**, the instant destruction in the outer peripheral region **120** becomes easier to prevent, as described below. Instant destruction is, for example, a destruction of an element before a current is shut down at turn-off.
(141) For example, if the acceleration energy of proton is 300 keV, the integral value of doping concentration of the collector region **22** is approximately 1.00E+14 [cm.sup.−2], which results in the maximum value of clamping energy. In this case, for example, the proton range of the first peak P**1** becomes 3.13 μm. Moreover, 1.00E+14 [cm.sup.−2] indicates 1.00×10.sup.14 [cm.sup.−2].
(142) Also, if the acceleration energy of proton is 400 keV, the integral value of doping concentration of the collector region **22** is approximately 1.00E+13 [cm.sup.−2], which results in the maximum value of clamping energy. In this case, for example, the proton range becomes 4.52 μm.
(143) Moreover, the integral value of doping concentration of the collector region **22** is substantially equal to the implanted dosage of back surface boron, and thus the implanted dosage may be set to be the integral value of doping concentration of the collector region **22**.
(144) The semiconductor device **100** of the present example makes the clamping breakdown voltage of the active region **110** smaller than the clamping breakdown voltage of the outer peripheral region **120** by controlling the integral concentration of the collector region **22** and the peak depth of the field stop region **20**. This allows the clamping energy of the semiconductor device **100** to improve.
(145) FIG. **3**B shows a relation between the doping concentration of the collector region **22** and the breakdown voltage of an element. The vertical axis indicates the breakdown voltage of an element, and the horizontal axis indicates the doping concentration of the collector region **22** [cm.sup.−3]. The collector region **22** includes a doping concentration distribution with peaks. In this case, the doping concentration of the collector region **22** may be a Peak value of the doping concentration of the collector region **22**. In the present example, a case where the field stop region **20** has four peaks is described.
(146) The graph G**1** indicates the active model and edge model when the field stop region **20** has four peaks. The active model and edge model is a simulation model that accounts for both the active region **110** and the outer peripheral region **120**. The graph G**2** indicates the active model when the field stop region **20** has four peaks. The active model is a simulation model that does not account for the outer peripheral region **120** but accounts only for the active region **110**.
(147) By decreasing the doping concentration of the collector region **22**, the breakdown voltage of the active region **110** falls lower than the breakdown voltage of the outer peripheral region **120**. That is, the breakdown voltage of an element becomes the breakdown voltage of the active region **110**. In this manner, when the breakdown voltage of an element starts to be determined in the active region **110**, clamping energy increases.
(148) On the other hand, by increasing the doping concentration of the collector region **22**, the breakdown voltage of the active region **110** increases greater than the breakdown voltage of the outer peripheral region **120**. That is, the breakdown voltage of an element becomes the breakdown voltage of the outer peripheral region **120**. In this manner, when the breakdown voltage of an element starts to be determined in the outer peripheral region **120**, clamping energy falls. Accordingly, in order to raise clamping energy, the doping concentration of the collector region **22** may be 6×10.sup.17 (/cm.sup.3) or less, or may be 5×10.sup.17 (/cm.sup.3) or less.
(149) FIG. **4**A shows an example of the current/voltage characteristic of the semiconductor device **100** according to the embodiment example. The vertical axis indicates a collector-emitter current

Ice [A] and a collector-emitter voltage Vce [V], and the horizontal axis indicates time [s]. Also, in a sectional view of the semiconductor device **100**, regions where electron current density is high are indicated by broken lines at the time T**1** and the time T**2** respectively. The doping concentration of the collector region **22** of the present example, in FIG. **3**B, is set to be within a range where the clamping breakdown voltage is determined in the active region **110**.
(150) Time T**1** is a time when the semiconductor device **100** turns off raising a collector-emitter voltage Vce. At the time T**1**, electron current is concentrated in the region indicated by the broken line of the active region **110**. That is, a part of the region indicated by the broken line of the active region **110** forms the peak regions of electron current density.
(151) Time T**2** indicates a time when a collector-emitter voltage Vce rises and is clamped around 800V. At the time T**2**, the temperature of a part where the avalanche breakdown occurs rises. Raised temperature intensifies lattice vibration, scattering electrons. If a current becomes less easy to flow due to the scattering of electrons, the peak regions of electron current density may move from the active region **110** toward an end portion of the side of the outer peripheral region **120** of the well region **17**.
(152) In this manner, if the breakdown voltage of the active region **110** is smaller than the breakdown voltage of the outer peripheral region **120**, the peak regions of electron current density formed on the side of the active region **110** move to the side of the outer peripheral region **120**. This temporarily lowers the temperature of the active region **110**. Also, if the peak regions of electron current density by an avalanche breakdown move to the side of the outer peripheral region **120**, power (that is, Ice×Vce) falls. Accordingly, the raised temperature of the semiconductor device **100** can be reduced so as to prevent instant destruction.
(153) FIG. **4**B shows an example of the current/voltage characteristic of the semiconductor device **500** according to the comparative example. The vertical axis indicates a collector-emitter current Ice [A] and a collector-emitter voltage Vce [V], and the horizontal axis indicates time [s]. Also, in the present example, the peak regions of electron current density are indicated by broken lines at the time T**3** and the time T**4** respectively. The doping concentration of the collector region **22** of the present example, in FIG. **3**B, is set to be within a range where the clamping breakdown voltage is determined in the outer peripheral region **120**.
(154) The semiconductor device **500** has the integral concentration of the collector region **22** set so that the breakdown voltage of the outer peripheral region **120** is smaller than the breakdown voltage of the active region **110**. Turn-offs cause an avalanche breakdown to occur on the side of the outer peripheral region **120** of the well region **17**, forming the peak regions of electron current density in the part indicated by the broken line. From time T**3** to time T**4**, power is high, and the peak regions of electron current density remain in the position indicated by the broken line. Therefore, a current crowding causes the temperature to continue to rise. In this manner, in the semiconductor device **500**, an element becomes easier to be disrupted, and the clamping overload cannot improve.
(155) In the outer peripheral region **120**, an avalanche breakdown tends to occur because carriers tend to concentrate in a contact hole edge of the side of the outer peripheral region **120** in addition to the amplified avalanche breakdown according to the hole injection efficiency of the collector region **22**. A rise in concentration of holes on the side of the back surface **23** amplifies an avalanche breakdown at clamping.
(156) The semiconductor device **100** of the present example decreases the hole injection efficiency of the collector region **22** by setting the depth of the first peak P**1** of the doping concentration of the collector region **22** at an appropriate depth, so as to make the clamping breakdown voltage of the active region **110** smaller than the clamping breakdown voltage of the outer peripheral region **120**. This allows an instant destruction to be suppressed because an avalanche breakdown moves to the outer peripheral region **120** after an avalanche breakdown occurred in the active region **110**. The doping concentration of the collector region **22** and the depth of the first peak P**1** may be set so that

clamping energy becomes maximum. An appropriate range of the doping concentration of the collector region **22** and the depth of the first peak P**1** will be described below.
(157) Moreover, if it is designed to have a breakdown voltage greater than the voltage overshoot (which is a peak value of voltage) estimated for preventing element destruction when shutting down a current, it is required to increase the thickness in the depth direction of the semiconductor substrate **10** or to increase the area of the outer peripheral region **120**. In order to reduce the chip cost, the breakdown voltage of an element may be designed to be a voltage overshoot or less. The semiconductor device **100** of the present example allows element destruction to be prevented while reducing the chip cost, by causing an avalanche breakdown to occur in the active region **110** at turn-off.
(158) Also, the total dosage of the field stop region **20** and the total dosage of the collector region **22** may be determined so that the breakdown voltage of the active region **110** becomes smaller than the breakdown voltage of the outer peripheral region **120**. For example, the total dosage of the field stop region **20** may be 10-fold or less of the total dosage of the collector region **22**, or may be 5-fold or less of the total dosage of the collector region **22**.
(159) FIG. **5**A shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**. The vertical axis indicates the depth of the first peak P**1**, y**1** [μm], and the horizontal axis indicates the integral concentration of the collector region **22**, x [cm.sup.−2].
(160) Here, the following base line L**1** is calculated based on the acceleration energy in the shallowest part of proton obtained in FIG. **3**A and the data at the two points where clamping energy becomes maximum.
Base line $L1$: $y1=(-6.0367E-01)\ln(x)+(2.2590E+01)$
That is, the base line L**1** indicates a relation between the integral concentration of the collector region **22** in which clamping energy becomes maximum and the depth of the first peak P**1** in which clamping energy becomes maximum.
(161) Region R**1** indicates the region within the range of ±15% of the base line L**1**. The region R**1** of the present example is a region between the line A**1** and the line B**1**. The line A**1** and the line B**1** of the present example are represented by the following equation.
Line $A1$: $y1=(-7.4699E-01)\ln(x)+(2.7810E+01)$
Line $B1$: $y1=(-4.7772E-01)\ln(x)+(1.7960E+01)$
(162) In this case, the depth of the first peak P**1** and the integral concentration of the collector region **22** belong to the region R**1**. If in the region R**1**, the hole injection efficiency of the collector region **22** can be suppressed so as to make the clamping breakdown voltage of the outer peripheral region **120** smaller than the clamping breakdown voltage of the active region **110**. The integral concentration of the collector region **22** may be 1.00E16 cm.sup.−2 or less, or may be 8.00E15 cm.sup.−2 or less. The depth of the first peak P**1** may be 0.5 μm or more and 7.2 μm or less.
(163) The semiconductor device **100** of the present example can cause an avalanche breakdown at clamping to occur on the side of the active region **110**, by setting the integral concentration of the collector region **22** and the depth of the first peak P**1** to be in the region R**1**. This improves a clamping breakdown voltage.
(164) FIG. **5**B shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**. The region R**1** of the present example indicates a region corresponding to ±10% of the base line L**1**.
(165) The base line L**1** is identical to the base line L**1** in FIG. **5**A. The line A**1** and the line B**1** of the present example are represented by the following equation.
Line $A1$: $y1=(-6.9487E-01)\ln(x)+(2.5930E+01)$
Line $B1$: $y1=(-5.2115E-01)\ln(x)+(1.9540E+01)$
(166) Similarly, in this case, the depth of the first peak P**1** and the integral concentration of the collector region **22** may be set to belong to the region R**1**.

(167) FIG. **5**C shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**. The region R**1** of the present example indicates a region corresponding to ±5% of the base line L**1**.
(168) The base line L**1** is identical to the base line L**1** in FIG. **5**A. The line A**1** and the line B**1** of the present example are represented by the following equation.
Line $A1$: $y1=(-6.4710E-01)\ln(x)+(2.4190E+01)$
Line $B1$: $y1=(-5.6458E-01)\ln(x)+(2.1130E+01)$
(169) Similarly, in this case, the depth of the first peak P**1** and the integral concentration of the collector region **22** may be set to belong to the region R**1** of the present example.
(170) FIG. **6**A shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**. The base line L**1**, the line A, and the line B are identical to each line in FIG. **5**A. That is, it indicates a region corresponding to ±15% of the base line L**1**. However, the region R**1** of the present example indicates a region where the integral concentration of the collector region **22** is 3.00E14 cm.sup.−2 or less. The depth of the first peak P**1** may be 2.0 μm or more and 7.2 μm or less. The depth of the first peak P**1** and the integral concentration of the collector region **22** may be set to belong to the region R**1** of the present example.
(171) FIG. **6**B shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**. The base line L**1**, the line A, and the line B are identical to each line in FIG. **5**A. That is, it indicates a region corresponding to ±15% of the base line L**1**. However, the region R**1** of the present example indicates a region where the integral concentration of the collector region **22** is 2.00E14 cm.sup.−2 or less. The depth of the first peak P**1** may be 2.2 μm or more and 7.2 μm or less. The depth of the first peak P**1** and the integral concentration of the collector region **22** may be set to belong to the region R**1** of the present example.
(172) FIG. **6**C shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**. The base line L**1**, the line A, and the line B are identical to each line in FIG. **5**A. That is, it indicates a region corresponding to ±15% of the base line L**1**. However, the region R**1** of the present example indicates a region where the integral concentration of the collector region **22** is 1.00E14 cm.sup.−2 or less. The depth of the first peak P**1** may be 2.5 μm or more and 7.2 μm or less. The depth of the first peak P**1** and the integral concentration of the collector region **22** may be set to belong to the region R**1** of the present example.
(173) FIG. **6**D shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**. The base line L**1**, the line A, and the line B are identical to each line in FIG. **5**A. That is, it indicates a region corresponding to ±15% of the base line L**1**. However, the region R**1** of the present example indicates a region where the integral concentration of the collector region **22** is 5.00E13 cm.sup.−2 or less. The depth of the first peak P**1** may be 3.0 μm or more and 7.2 μm or less. The depth of the first peak P**1** and the integral concentration of the collector region **22** may be set to belong to the region R**1** of the present example.
(174) FIG. **6**E shows another example of a relation between the integral concentration of the collector region **22** and the depth of the first peak P**1**. The base line L**1**, the line A, and the line B are identical to each line in FIG. **5**A. That is, it indicates a region corresponding to ±15% of the base line L**1**. However, the region R**1** of the present example indicates a region where the integral concentration of the collector region **22** is 3.00E13 cm.sup.−2 or less. The depth of the first peak P**1** may be 3.2 μm or more and 7.2 μm or less. The depth of the first peak P**1** and the integral concentration of the collector region **22** may be set to belong to the region R**1** of the present example.
(175) FIG. **6**F shows another example of a relation between the integral concentration of the

collector region **22** and the depth of the first peak P**1**. The base line L**1**, the line A, and the line B are identical to each line in FIG. **5**A. That is, it indicates a region corresponding to ±15% of the base line L**1**. However, the region R**1** of the present example indicates a region where the integral concentration of the collector region **22** is 1.00E13 cm.sup.−2 or less. The depth of the first peak P**1** may be 3.6 μm or more and 7.2 μm or less. The depth of the first peak P**1** and the integral concentration of the collector region **22** may be set to belong to the region R**1** of the present example.

(176) Moreover, also for FIG. **5**B and FIG. **5**C, the range of an integral concentration of the collector region **22** may be restricted similarly to FIG. **6**A to FIG. **6**F. For example, in FIG. **5**B and FIG. **5**C, the region R**1** may be a region with the integral concentration of the collector region **22** of 1.00E14 cm.sup.−2 or less, a region with that of the 5.00E13 cm.sup.−2 or less, a region with that of 3.00E13 cm.sup.−2 or less, or a region with that of 1.00E13 cm.sup.−2 or less.

(177) FIG. **7**A shows another example of a relation between the integral concentration of the collector region **22** and the depth of a second peak P**2**. The vertical axis indicates the depth of the second peak P**2**, y**2** [μm], and the horizontal axis indicates the integral concentration of the collector region **22**, x [cm.sup.−2].

(178) Here, the following base line L**2** is calculated based on the acceleration energy in the shallowest part of proton obtained in FIG. **3**A and the data at the two points where clamping energy becomes maximum.

Base line $L2$: $y2=(-2.4885E+00)\ln(x)+(9.1580E+01)$

(179) That is, the base line L**2** indicates a relation between the integral concentration of the collector region **22** in which clamping energy becomes maximum and the depth of the second peak P**2** in which clamping energy becomes maximum.

(180) Region R**2** indicates the region within the range of ±15% of the base line L**2**. The region R**2** of the present example is a region between the line A**2** and the line B**2**. The line A**2** and the line B**2** of the present example are represented by the following equation.

Line $A2$: $y2=(-3.1095E+00)\ln(x)+(1.1416E+02)$

Line $B2$: $y2=(-1.9239E+00)\ln(x)+(7.1030E+01)$

(181) In this case, the depth of the second peak P**2** and the integral concentration of the collector region **22** belong to the region R**2**. If in the region R**2**, clamping energy becomes sufficiently high. The depth of the second peak P**2** may be 3.5 μm or more and 28 μm or less.

(182) FIG. **7**B shows another example of a relation between the integral concentration of the collector region **22** and the depth of the second peak P**2**. The region R**2** of the present example indicates a region corresponding to ±10% of the base line L**2**.

(183) The base line L**2** is identical to the base line L**2** in FIG. **7**A. The line A**2** and the line B**2** of the present example are represented by the following equation.

Line $A2$: $y2=(-2.8924E+00)\ln(x)+(1.0629E+02)$

Line $B2$: $y2=(-2.1020E+00)\ln(x)+(7.7530E+01)$

(184) Similarly, in this case, the depth of the second peak P**2** and the integral concentration of the collector region **22** may be set to belong to the region R**2**.

(185) FIG. **7**C shows another example of a relation between the integral concentration of the collector region **22** and the depth of the second peak P**2**. The region R**2** of the present example indicates a region corresponding to ±5% of the base line L**2**.

(186) The base line L**2** is identical to the base line L**2** in FIG. **7**A. The line A**2** and the line B**2** of the present example are represented by the following equation.

Line $A2$: $y2=(-2.4885E+00)\ln(x)+(9.1580E+01)$

Line $B2$: $y2=(-2.2931E+00)\ln(x)+(8.4470E+01)$

(187) Similarly, in this case, the depth of the second peak P**2** and the integral concentration of the collector region **22** may be set to belong to the region R**2**.

(188) Moreover, also for FIG. **7**A to FIG. **7**C, the range of an integral concentration of the collector

region **22** may be restricted similarly to FIG. **6**A to FIG. **6**F. For example, in FIG. **7**A to FIG. **7**C, the region R**2** may be a region with the integral concentration of the collector region **22** of 1.00E14 cm.sup.−2 or less, a region with that of the 5.00E13 cm.sup.−2 or less, a region with that of 3.00E13 cm.sup.−2 or less, or a region with that of 1.00E13 cm.sup.−2 or less.
(189) FIG. **8**A is a diagram for describing a relation between a float inductance Ls and a collector current decrease rate dIce/dt. The float inductance Ls is set to be Xc [nH], and the collector current decrease rate dIce/dt is set to be Yc [A/μs]. The line C**1** is represented by the following equation.
Line $C1$: $Yc=10000Xc$.sup.−1
(190) The float inductance Ls and the collector current decrease rate dIce/dt is set to the range that is greater than the line C**1**. The range that is greater than the line C**1** refers to a region that is located above the line C**1** in the graph indicating a relation between the float inductance Ls and the collector current decrease rate dIce/dt. The range that is greater than the line C**1** is a region filled in with a pattern.
(191) The float inductance Ls and the collector current decrease rate dIce/dt may be set to the range that is greater than the line C**2**. The line C**2** is represented by the following equation.
Line $C2$: $Yc=20000Xc$.sup.−1
(192) The float inductance Ls and the collector current decrease rate dIce/dt may be set to the range that is greater than the line C**3**. The line C**3** is represented by the following equation.
Line $C3$: $Yc=50000Xc$.sup.−1
(193) Here, the greater the float inductance Ls becomes, the easier it becomes to maintain a current, thereby a voltage overshoot increases when shutting down a current. Also, the greater the collector current decrease rate dIce/dt is, the greater the voltage overshoot becomes when shutting down the current. Therefore, if the float inductance Ls and the collector current decrease rate dIce/dt are set to be in the range that is greater than any one of the line C**1** to the line C**3**, the voltage overshoot tends to snap off. The semiconductor device **100** of the present example can prevent instant destruction even if the float inductance Ls and the collector current decrease rate dIce/dt are set to be in the range that is greater than any one of the line C**1** to the line C**3**, the voltage overshoot tends to snap off.
(194) FIG. **8**B is a diagram for describing a relation between comparative stray inductance Ls.Math.A and current density dJce/dt. The comparative stray inductance Ls.Math.A is set to be Xd [nH cm.sup.2], and the current density dJce/dt is set to be Yd [A/(cm.sup.2 μs)].
(195) The comparative stray inductance Ls.Math.A is a specific value that results from multiplying the float inductance Ls by the area A of the active region **110** (cm.sup.2). The current density dJce/dt is a value that results from dividing the collector current decrease rate dIce/dt by the area A of the active region **110** (cm.sup.2).
(196) The comparative stray inductance Ls.Math.A and the current density dJce/dt is set to the range that is greater than the line D**1**. The line D**1** is represented by the following equation.
Line $D1$: $Yd=10000Xd$.sup.−1
(197) The comparative stray inductance Ls.Math.A and the current density dJce/dt may be set to the range that is greater than the line D**2**. The line D**2** is represented by the following equation.
Line $D2$: $Yd=20000Xd$.sup.−1
(198) The comparative stray inductance Ls.Math.A and the current density dJce/dt may be set to the range that is greater than the line D**3**. The line D**3** is represented by the following equation.
Line $D3$: $Yd=50000Xd$.sup.−1
(199) Here, the greater the comparative stray inductance Ls.Math.A or the current density dJce/dt are, the greater the voltage overshoot becomes when shutting down a current. Therefore, if the comparative stray inductance Ls.Math.A or the current density dJce/dt are set to be in the range that is any one or above of the line D**1** to the line D**3**, the voltage overshoot tends to snap off. The semiconductor device **100** of the present example can prevent instant destruction even if the comparative stray inductance Ls.Math.A or the current density dJce/dt are set to be in the range that

is greater than any one of the line D**1** to the line D**3**, the voltage overshoot tends to snap off.
(200) FIG. **9** is a diagram for describing a relation between the comparative stray inductance Ls.Math.A and the comparative gate resistance Rg.Math.A. The comparative stray inductance Ls.Math.A is set to be Xe [nH cm.sup.2], the comparative gate resistance Rg.Math.A is set to be Ye [Ωcm.sup.2]. The comparative gate resistance Rg.Math.A is a specific value that results from multiplying the gate resistance Rg of a driving circuit of the semiconductor device **100** by the area A of the active region **110** (cm.sup.2).
(201) The collector current decrease rate dIce/dt is changed by the gate resistance Rg of a driving circuit of the semiconductor device **100**, in addition to the float inductance Ls. If the gate resistance Rg decreases, the collector current decrease rate dIce/dt tends to increase.
(202) The comparative stray inductance Ls.Math.A and the comparative gate resistance Rg.Math.A may be set to be the range of the line E**1** or less. The line E**1** is represented by the following equation.
Line $E1$: $Ye=(4.000\mathrm{E}{-}01)Xe$
(203) The comparative stray inductance Ls.Math.A and the comparative gate resistance Rg.Math.A may be set to be the range of the line E**2** or less. The line E**2** is represented by the following equation.
Line $E2$: $Ye=(2.000\mathrm{E}{-}01)Xe$
(204) The comparative stray inductance Ls.Math.A and the comparative gate resistance Rg.Math.A may be set to be the range of the line E**3** or less. The line E**3** is represented by the following equation.
Line $E3$: $Ye=(8.000\mathrm{E}{-}02)Xe$
(205) Here, the smaller the comparative gate resistance Rg.Math.A is, the greater the voltage overshoot becomes when shutting down a current. Therefore, if the comparative stray inductance Ls.Math.A and the comparative gate resistance Rg.Math.A are set to be in the range that is any one of or below the line E**1** to the line E**3**, the voltage overshoot tends to snap off. The semiconductor device **100** of the present example can prevent instant destruction even if the comparative stray inductance Ls.Math.A and the comparative gate resistance Rg.Math.A are set to be in the range that is any one of or below the line E**1** to the line E**3**.
(206) While the embodiments of the present invention have been described, the technical scope of the present invention is not limited to the above described embodiments. It is apparent to persons skilled in the art that various alterations or improvements can be added to the above-described embodiments. It is also apparent from the scope of the claims that the embodiments added with such alterations or improvements can be included in the technical scope of the present invention.
(207) The operations, procedures, steps, and stages of each process performed by an apparatus, system, program, and method shown in the claims, embodiments, or diagrams can be performed in any order as long as the order is not indicated by “prior to,” “before,” or the like and as long as the output from a previous process is not used in a later process. Even if the process flow is described using phrases such as “first” or “next” in the claims, embodiments, or diagrams, it does not necessarily mean that the process must be performed in this order.
EXPLANATION OF REFERENCES
(208) **10** . . . Semiconductor substrate, **12** . . . Emitter region, **14** . . . Base region, **15** . . . Contact region, **16** . . . Accumulation region, **17** . . . Well region, **18** . . . Drift region, **20** . . . Field stop region, **21** . . . Front surface, **22** . . . Collector region, **23** . . . Back surface, **24** . . . Collector electrode, **25** . . . Connection portion, **30** . . . Dummy trench portion, **31** . . . Extending portion, **32** . . . Dummy dielectric film, **33** . . . Connection section, **34** . . . Dummy conductive portion, **38** . . . Interlayer dielectric film, **39** . . . Oxide film, **40** . . . Gate trench portion, **41** . . . Extending portion, **42** . . . Gate dielectric film, **43** . . . Connection section, **44** . . . Gate conductive portion, **50** . . . Gate metal layer, **51** . . . Gate runner, **52** . . . Emitter electrode, **54** . . . Contact hole, **55** . . . Contact hole, **56** . . . Contact hole, **60** . . . Dummy trench region, **61** . . . Mesa portion, **65** . . . Well-contact region,

70 . . . Transistor portion, **71** . . . Mesa portion, **76** . . . Temperature sensing portion, **80** . . . Diode portion, **81** . . . Mesa portion, **82** . . . Cathode region, **90** . . . Boundary portion, **91** . . . Mesa portion, **92** . . . Guard ring portion, **94** . . . Electrode layer, **96** . . . Channel stopper region, **100** . . . Semiconductor device, **110** . . . Active region, **120** . . . Outer peripheral region, **130** . . . Pad region, **131** . . . Gate pad, **132** . . . Sense-IGBT, **133** . . . Sense-emitter pad, **134** . . . Anode pad, **135** . . . Cathode pad, **500** . . . Semiconductor device

Claims

1. A semiconductor device comprising: a drift region of a first conductivity type provided on a semiconductor substrate; a field stop region of a first conductivity type provided below the drift region and having one or more peaks of a doping concentration in a depth direction; and a collector region of a second conductivity type provided below the field stop region, wherein when an integral in the depth direction of a dopant concentration of the collector region is set to be x [cm.sup.−2], a depth of a first peak of the one or more peaks of the doping concentration of the field stop region that is a shallowest from a back surface of the semiconductor substrate is set to be y**1** [μm],
line $A1{:}y1[\mu m]=(-7.4699E\text{-}01)\ln(x\,[cm.sup.-2])+(2.7810E+01)$, and
line $B1{:}y1[\mu m]=(-4.7772E\text{-}01)\ln(x\,[cm.sup.-2])+(1.7960E+01)$, the depth of the first peak [μm] and the integral in the depth direction of the dopant concentration [cm.sup.−2] of the collector region are within a range between the line A**1** [μm/cm.sup.−2] and the line B**1** [μm/cm.sup.−2], and the collector region includes a peak of the dopant concentration in the depth direction that is higher than a doping concentration of the first peak in the field stop region.
2. The semiconductor device according to claim 1, wherein the integral of concentration of the collector region is 8.00E15 cm.sup.−2 or less.
3. The semiconductor device according to claim 1, wherein the integral of concentration of the collector region is 3.00E14 cm.sup.−2 or less.
4. The semiconductor device according to claim 1, wherein the integral of concentration of the collector region is 2.00E14 cm.sup.−2 or less.
5. The semiconductor device according to claim 1, wherein the integral of concentration of the collector region is 1.00E14 cm.sup.−2 or less.
6. The semiconductor device according to claim 1, wherein the integral of concentration of the collector region is 5.00E13 cm.sup.−2 or less.
7. The semiconductor device according to claim 1, wherein the integral of concentration of the collector region is 3.00E13 cm.sup.−2 or less.
8. The semiconductor device according to claim 1, wherein the integral of concentration of the collector region is 1.00E13 cm.sup.−2 or less.
9. The semiconductor device according to claim 1, wherein a depth of the first peak is 0.5 μm or more and 7.2 μm or less.
10. The semiconductor device according to claim 1, wherein a depth of the first peak is 2.0 μm or more and 7.2 μm or less.
11. The semiconductor device according to claim 1, wherein a depth of a second peak of the one or more peaks of the doping concentration that is a second shallowest from the back surface is set to be y**2** [μm],
line $A2{:}y2[\mu m]=(-3.1095E+00)\ln(x[cm.sup.-2])+(1.1416E+02)$, and
line $B2{:}y2[\mu m]=(-1.9239E+00)\ln(x[cm.sup.-2])+(7.1030E+01)$, a depth of the second peak and the integral of concentration are within a range between the line A**2** and the line B**2**.
12. The semiconductor device according to claim 11, wherein a depth of the second peak is 3.5 μm or more and 28 μm or less.
13. The semiconductor device according to claim 1, wherein a dopant of the one or more peaks of

the doping concentration is hydrogen.
14. The semiconductor device according to claim 1, further comprising: an active region provided on the semiconductor substrate; and an outer peripheral region provided in an outer periphery of the active region in a top view of the semiconductor substrate.
15. The semiconductor device according to claim 1, further comprising: a base region of the second conductivity type provided above the drift region; an emitter region of the first conductivity type provided above the base region and whose doping concentration is higher than that of the drift region; a contact region of the second conductivity type provided above the base region and whose doping concentration is higher than that of the base region; and a plurality of gate trench portions provided on the semiconductor substrate.
16. The semiconductor device according to claim 1, wherein a doping concentration in a boundary between the field stop region and the collector region is 1E16 cm.sup.−3 or less.
17. The semiconductor device according to claim 1, wherein a doping concentration in a boundary between the field stop region and the collector region is 5E15 cm−3 or less.
18. The semiconductor device according to claim 1, wherein a doping concentration in a boundary between the field stop region and the collector region is 2E15 cm.sup.−3 or less.
19. The semiconductor device according to claim 1, wherein a hydrogen chemical concentration in a boundary between the field stop region and the collector region is 1E18 cm.sup.−3 or less.
20. The semiconductor device according to claim 1, wherein a hydrogen chemical concentration in a boundary between the field stop region and the collector region is 1E17 cm.sup.−3 or less.
21. The semiconductor device according to claim 1, wherein a hydrogen chemical concentration in a boundary between the field stop region and the collector region is 1E15 cm.sup.−3 or more.
22. The semiconductor device according to claim 1, wherein a hydrogen chemical concentration in a boundary between the field stop region and the collector region is 1E16 cm.sup.−3 or more.
23. The semiconductor device according to claim 1, wherein a dopant of the first peak is phosphorous.
24. The semiconductor device according to claim 1, wherein a dopant of peaks of the doping concentration other than the first peak out of the one or more peaks is hydrogen.
25. The semiconductor device according to claim 1, wherein a front surface side region of the semiconductor substrate does not comprise a lifetime killer region.
26. A system comprising the semiconductor device according to claim 1, wherein the semiconductor device is coupled to a circuit having a stray inductance Ls set to be Xc [nH], and a collector current decrease rate dice/dt set to be Yc [A/μs], and
line $C1{:}Yc=10000Xc$.sup.−1, the stray inductance Ls and the collector current decrease rate dice/dt are within a range that is greater than the line C**1**.